

§ 11.5 Alternating Series and Absolute Convergence

(교대급수와 절대수렴)

Ex.8 까지

■ Alternating Series (교대급수)

An **alternating series** is a series whose terms are alternately positive and negative.

(교대급수 판정법)

Alternating Series Test If the alternating series

$$\sum_{n=1}^{\infty} (-1)^{n-1} b_n = b_1 - b_2 + b_3 - b_4 + b_5 - b_6 + \cdots \quad (b_n > 0)$$

satisfies the conditions

(i) $b_{n+1} \leq b_n$ for all n

(ii) $\lim_{n \rightarrow \infty} b_n = 0$

then the series is convergent.

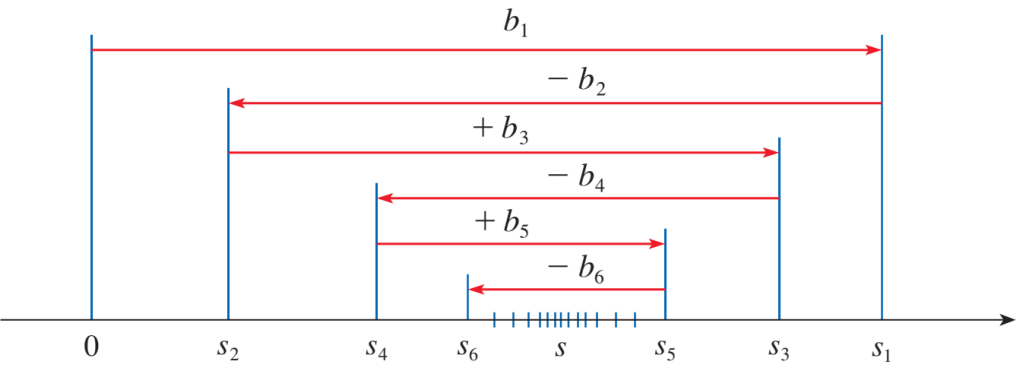


Figure 1

EXAMPLE 1 The alternating harmonic series

$$1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \dots = \sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n}$$

EXAMPLE 2 The series $\sum_{n=1}^{\infty} \frac{(-1)^n 3n}{4n - 1}$

EXAMPLE 3 Test the series $\sum_{n=1}^{\infty} (-1)^{n+1} \frac{n^2}{n^3 + 1}$ for convergence or divergence.

Solution:

■ Estimating Sums of Alternating Series (교대급수 합의 추정)

Alternating Series Estimation Theorem If $s = \sum (-1)^{n-1} b_n$, where $b_n > 0$, is the sum of an alternating series that satisfies

$$(i) \ b_{n+1} \leq b_n \quad \text{and} \quad (ii) \ \lim_{n \rightarrow \infty} b_n = 0$$

then

$$|R_n| = |s - s_n| \leq b_{n+1}$$

EXAMPLE 4 Find the sum of the series $\sum_{n=0}^{\infty} \frac{(-1)^n}{n!}$ correct to three decimal places.

Solution:

■ Absolute Convergence and Conditional Convergence (절대수렴과 조건부수렴)

Given any series $\sum a_n$, we can consider the corresponding series

$$|a_n| = |a_1| + |a_2| + |a_3| + \cdots$$

whose terms are the absolute values of the terms of the original series.

1 Definition A series $\sum a_n$ is called **absolutely convergent** if the series of absolute values $\sum |a_n|$ is convergent.

EXAMPLE 5 The alternating series

$$\sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n^2} = 1 - \frac{1}{2^2} + \frac{1}{3^2} - \frac{1}{4^2} + \cdots$$

2 Definition A series $\sum a_n$ is called **conditionally convergent** if it is convergent but not absolutely convergent; that is, if $\sum a_n$ converges but $\sum |a_n|$ diverges.

EXAMPLE 6 $\sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n} = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \dots$

3 Theorem If a series $\sum a_n$ is absolutely convergent, then it is convergent.

EXAMPLE 7 Determine whether the series

$$\sum_{n=1}^{\infty} \frac{\cos n}{n^2} = \frac{\cos 1}{1^2} + \frac{\cos 2}{2^2} + \frac{\cos 3}{3^2} + \dots$$

is convergent or divergent.

Solution:

EXAMPLE 8 Determine whether the series is absolutely convergent, conditionally convergent, or divergent.

$$(a) \sum_{n=1}^{\infty} \frac{(-1)^n}{n^3} \quad (b) \sum_{n=1}^{\infty} \frac{(-1)^n}{\sqrt[3]{n}} \quad (c) \sum_{n=1}^{\infty} (-1)^n \frac{n}{2n+1}$$

Solution:

